

CS536 Final Project

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1 Introduction

Paper of interest:

- T. Sandnes, B. Grimstad, and O. Kolbjørnsen, “Multi-task neural networks by learned contextual inputs,” *Neural Networks*, vol. 179, p. 106528, Nov. 2024, doi: 10.1016/j.neunet.2024.106528.
- Preprint link: <https://arxiv.org/abs/2303.00788v2>
- Code from the original paper is not available.

Multi-task learning problem. Consider a set of regression tasks j . The aim is to learn the set of task-specific functions f_j mapping x_{ij} to y_{ij} . For a concrete example, consider the NCD Risk Factor Collaboration Height dataset of the mean height by age, sex, and year across 200 countries. Each country defines a separate task. We could train separate models for each country to predict height (y_{ij}) from age, sex, and year (x_{ij}). However, our model might perform better if we share knowledge across countries. The paper explores methods for multi-task learning that include the task as a contextual input to the model.

Theoretical background. The authors prove a “universal approximation theorem” for learned-context neural networks: that learned-context neural networks can achieve any task adaption if the network is sufficiently wide and deep, even if the task parameter is a scalar (one dimension). However, they note that a one-dimensional parameter is not ideal in practice, because it won’t be able to learn latent properties of the tasks.

Project aims. Here, I re-implement the learned-context neural network described in the paper, as well as a few simpler models for comparison. To extend the original paper, I also implement skip-connections in the learned-context neural network, allowing deeper layers to directly access the learned context parameter. This was suggested but not implemented by the authors.

2 Methods

Models. I implemented the following models as described in the paper.

- Linear regression model without task information.
- Linear regression model for each task.
- Multilayer perceptron without task information.
- Context-sensitive network. The task is one-hot-encoded and appended to the input. h is a neural network parameterized by α . c_j is the task indicator variable.

$$f_j(x_{ij}) = h(z_{ij}, \alpha), z_{ij} = \begin{bmatrix} x_{ij} \\ c_j \end{bmatrix}$$

- Learned-context neural network. Instead of one-hot encoding, the task is represented as a learned context parameter β_j and appended to the input.

$$f_j(x_{ij}) = h(z_{ij}, \alpha), z_{ij} = \begin{bmatrix} x_{ij} \\ \beta_j \end{bmatrix}$$

- Learned-context neural network with skip connection. Added a skip connection from the output of the first layer to the input of the last layer.

Hyperparameters. The key hyperparameter for the learned-context neural network is the dimension of the learned context parameter. The hidden dimension, number of hidden layers, learning rate, batch size, and number of epochs are also hyperparameters. Changing the hyperparameters did not change the relative ranking of the results. Therefore, the hyperparameters were set to the same value for all runs for comparison (dimension of learned context parameter = 30, hidden dimension = 300, number of hidden layers = 3, learning rate = 1e-3, batch size = 50, number of epochs = 10).

Height dataset. The dataset is the NCD Risk Factor Collaboration Height dataset of mean height by age, sex, and year across 200 countries. Each country defines a separate task j . The target is height (y_{ij}) and the features are age, sex, and year (x_{ij}).

Parkinson’s dataset. This dataset is the Oxford Parkinson’s Disease Telemonitoring Dataset from UCI Machine Learning Repository. Each of 42 patients defines a separate task j . There are roughly 200 data points per patient. The 16 continuous features were extracted from biomedical voice measurements (x_{ij}). The target is total UPDRS, which is a disease symptom score (y_{ij}).

3 Results

Table 1. RMSD by dataset and method.

Dataset	Height	Parkinson’s
Linear regression	6.748	0.948
Per-task linear regression	4.874	0.315
Multilayer perceptron	4.890	0.983
Context-sensitive network	1.533	0.964
Learned-context network	0.790	1.005
with one dimension	1.517	NA
with skip connections	0.700	0.830

Height dataset analysis. The learned-context neural network outperformed all other methods. Adding skip connections to the learned-context neural network improved its performance even more. Interestingly, when the context embedding was reduced to one dimension, the performance of the model is comparable to the context-sensitive network, which uses a simple one-hot encoding of the task as input. Population-level linear regression performs the worst, while training a separate linear regression model for each task performs about the same as the simple multilayer perceptron.

Parkinson’s dataset analysis. Surprisingly, the best-performing model on this dataset was the per-task linear regression. The population-level linear regression model performs about as well as the neural networks. Adding skip connections did improve the performance of the learned-context network, though it still performed much worse than the per-task linear regression.

What is the model learning? Here, I attempt to interpret the meaning of the learned context parameters from the height dataset. The learned context parameters were extracted from the model after training and embedded into two dimensions by PCA (Figure 1). Each context parameter represents a country. The points are colored by the mean height of 19-year-old boys in the year 2019. Along the PC1 axis, the countries are clearly ordered according to height, indicating that the context parameters encode meaningful information about which countries have taller people on average. The model is likely also learning latent features of the dataset that are not apparent at a glance. I also examined the learned context parameters from the learned-context network with one-dimensional context parameters (Figure 2). Even though the representation is cruder and the performance worse, we can still clearly see a correlation between the context parameter value and height.

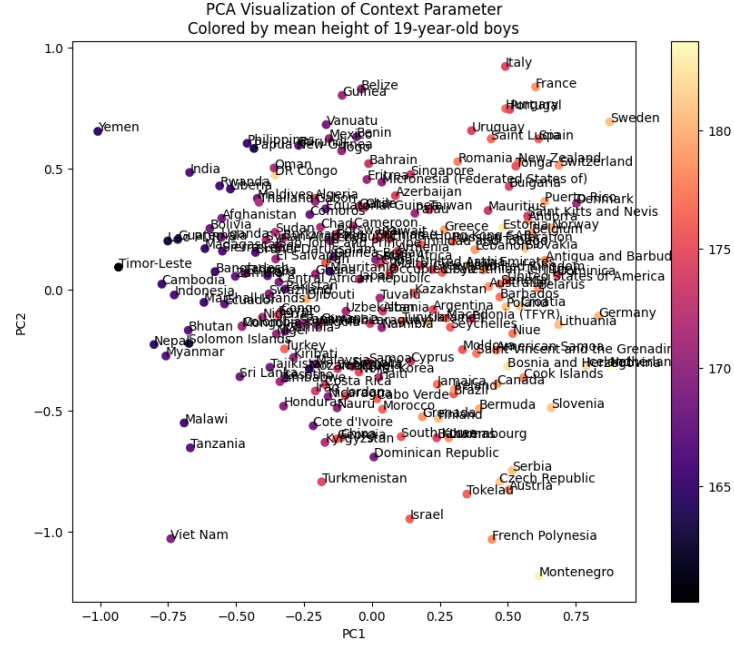


Figure 1: PCA visualization of context parameter

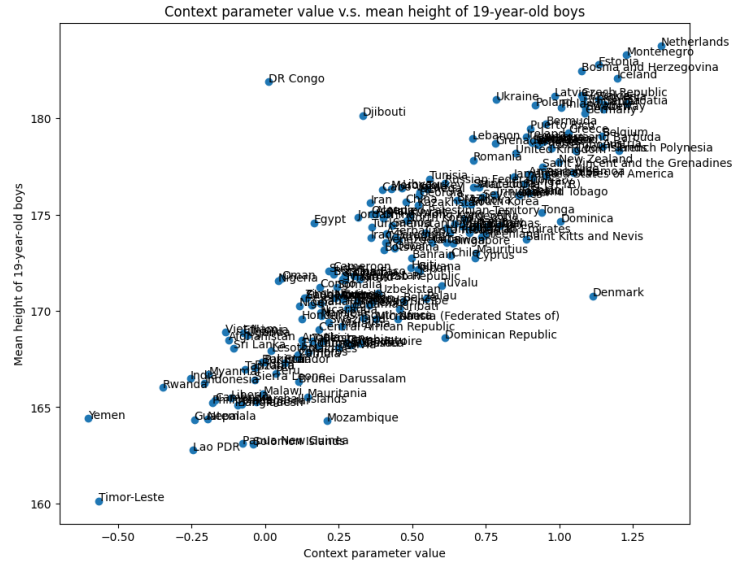


Figure 2: 1-dimensional context parameter vs. mean height of 19-year-old boys

4 Discussion

In summary, it is clear that learned context embeddings are a useful framework for multi-task learning. High-dimensional context embeddings can encode complex information about the task, allowing the model to learn latent patterns in the data. The context embeddings can also be interpreted to understand the relationships between tasks.

Extending the original paper, I show that implementing skip connections improves the performance of learned-context neural networks. Skip connections allow deeper layers to access task information directly from the initial layer, which prevents the model from "forgetting" information when there are many layers. Based on the results shown here, I recommend that future researchers training learned-context neural networks include skip connections in their architecture.

While learned-context neural networks are powerful, they are not appropriate for all use cases. Learned-context neural networks showed excellent performance over baselines on the height dataset, but performed poorly on the Parkinson's dataset compared to per-task linear regression. I suspect that the more complex models are overfitting to the training data, suggesting that a simpler model is best here. As mentioned earlier, the population-level linear regression model performs as well as the neural networks, while having the benefit of being more interpretable. Future work should explore how the performance of learned-context neural networks depends on the structure of the data.

5 References

- T. Sandnes, B. Grimstad, and O. Kolbjørnsen, "Multi-task neural networks by learned contextual inputs," *Neural Networks*, vol. 179, p. 106528, Nov. 2024, doi: 10.1016/j.neunet.2024.106528.
- NCD Risk Factor Collaboration, 2020. Height and body-mass index trajectories of school-aged children and adolescents from 1985 to 2019 in 200 countries and territories: a pooled analysis of 2181 population-based studies with 65 million participants. *The Lancet* 396, 1511–1524.
- Athanasios Tsanas, Max A. Little, Patrick E. McSharry, Lorraine O. Ramig (2009), 'Accurate telemonitoring of Parkinson's disease progression by non-invasive speech tests', *IEEE Transactions on Biomedical Engineering*.

6 Appendix

Attached with this writeup are the source code for `utils.py` and two Jupyter notebooks showing the analysis for each dataset. Code is also available at <https://github.com/alyssa-lee/learned-context-neural-networks>.